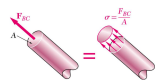


Normal Stress

Perpendicular to surface



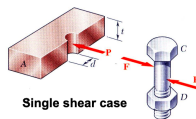
$$\sigma = \frac{F}{A}$$

Sign Convention

$\sigma > 0 \rightarrow$ Tension

$\sigma < 0 \rightarrow$ Compression

Bearing Stress

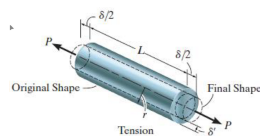


$$\sigma_b = \frac{P}{A_b} = \frac{P}{td}$$

$A_b \rightarrow$ projected area

$P \rightarrow$ bearing force

Poisson's RATIO



$$\nu = -\frac{\epsilon_{lat}}{\epsilon_{long}}$$

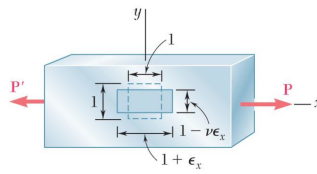
$$\epsilon_{lat} = \frac{\delta'}{R} \quad \epsilon_{long} = \frac{\delta}{L}$$

$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\nu \sigma_y}{E} - \frac{\nu \sigma_z}{E}$$

$$\epsilon_y = -\frac{\nu \sigma_x}{E} + \frac{\sigma_y}{E} - \frac{\nu \sigma_z}{E}$$

$$\epsilon_z = -\frac{\nu \sigma_x}{E} - \frac{\nu \sigma_y}{E} + \frac{\sigma_z}{E}$$

For isotropic bar $\sigma_y = \sigma_z = 0$



Slope after deformation

$$m' = m \cdot \frac{1 + \epsilon_y}{1 + \epsilon_x}$$

Torque

$$\tau_{max} = \frac{Tc}{J}$$

Angle of twist

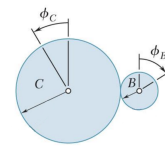
$$\phi = \frac{TL}{GJ}$$

Polar Moment of Inertia J

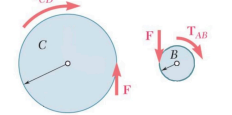
$$J = \frac{1}{2} \pi (c_2^4 - c_1^4)$$



Gears



$$r_B \phi_B = r_C \phi_C$$

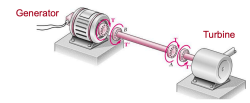


$$\sum m_C = 0 = -T_{CD} + r_C F$$

$$\sum m_B = 0 = -T_{AB} + r_B F$$

$$T_{AB} = T_{CD} \cdot \frac{r_B}{r_C}$$

Transmission Shafts



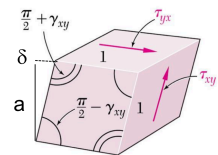
$$P = T\omega$$

Power with frequency

$$\omega = 2\pi f$$

$$P = 2\pi f T$$

Shear Strain



Shear sign conv

$$\gamma \approx \frac{\delta}{a}$$

$$\tau_{xy} = G\gamma_{xy}$$

$$\tau_{yz} = G\gamma_{yz}$$

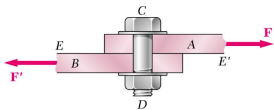
$$\tau_{zx} = G\gamma_{zx}$$

$$G = \frac{E}{2(1 + \nu)}$$

Shear Stress

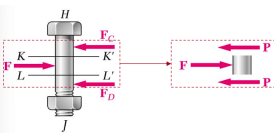
Parallel to surface

$$\tau_{ave} = \frac{P}{A} = \frac{F}{A}$$

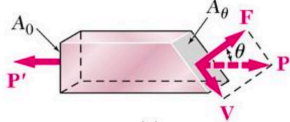


Double Shear

$$\tau_{ave} = \frac{P}{A} = \frac{F/2}{A} = \frac{F}{2A}$$



Stress on an Oblique Plane



$$F = P \cos(\theta) \quad V = P \sin(\theta)$$

$$A_0 = A_\theta \cos(\theta) \rightarrow A_\theta = \frac{A_0}{\cos(\theta)}$$

Average normal stress

$$\sigma = \frac{F}{A_\theta} = \frac{P}{A_0} \cos^2(\theta)$$

Average shear stress

$$\tau = \frac{V}{A_\theta} = \frac{P}{A_0} \sin(\theta) \cos(\theta)$$

Design Considerations

Largest load: P_u .

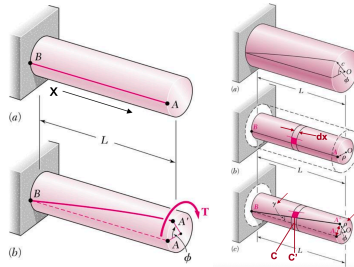
ultimate normal stress: $\sigma_u = \frac{P_u}{\sigma}$

ultimate shear stress $\tau_u = \frac{P_u}{A_\tau}$

Safety factor: $F.S.$

$$F.S. = \frac{P_u}{P_{all}} = \frac{\tau_u}{\tau_{all}} = \frac{\sigma_u}{\sigma_{all}}$$

Angle of Twist



$$\phi(x) = \frac{\phi x}{L}$$

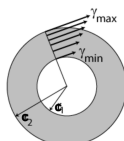
Shearing Strain

$$\gamma = \frac{\rho \phi}{L} \quad \gamma_{max} = \frac{c \phi}{L}$$

Shear strain at any distance

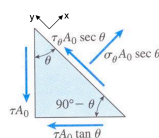
$$\gamma = \frac{\rho}{c} \gamma_{max}$$

Shearing strain on hollow tube



$$\gamma_{min} = \frac{c_1 \phi}{L} \quad \gamma_{max} = \frac{c_2 \phi}{L}$$

Inclined Plan



Sum forces in x dir: $\sigma_\theta = \tau \sin(2\theta)$

Sum forces in y dir: $\tau_\theta = \tau \cos(2\theta)$

$$\tau_{max} \text{ at } 0^\circ, \pm 90^\circ \quad \sigma_{max} \text{ at } \pm 45^\circ$$

$$\tau_{max} = \sigma_{max}$$

English: lb/in²

= psi

kip/in² = ksi

SI: N/m² = Pa

kN/m² = kPa

Bending Beams

Normal stress

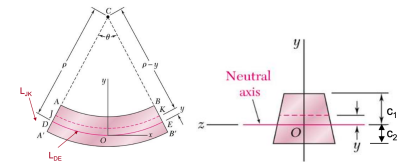
$$\sigma_{max} = -\frac{|M|_{max} c_1}{I}$$

$$\sigma_{max} = \frac{|M|_{max} c_2}{I}$$

$$= \frac{|M|_{max}}{S}$$

$$S_{min} = \frac{1}{6} b h^2$$

Pure Bending



$$\epsilon_x = -\frac{y}{\rho}$$

$$\leftrightarrow \epsilon_{max} = -c/\rho \quad \leftrightarrow \epsilon_{max} = c/\rho$$

$$\sigma_x = E \epsilon_x = -\frac{E y}{\rho} = -\frac{M y}{I}$$

Centroids for composite shapes

	Area	$\bar{y}$	Area $\times \bar{y}$
A_1	$b_1 h_1$	$0.5 h_1$	$0.5 h_1^2 b_1$
A_2	$b_2 h_2$	$h_1 + 0.5 h_2$	$h_2 b_2 (h_1 + 0.5 h_2)$
A_3	$b_3 h_3$	$h_1 + h_2 + 0.5 h_3$	$h_3 b_3 (h_1 + h_2 + 0.5 h_3)$
	$\sum A_i$		$\sum A_i \bar{y}_i$

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i}$$

Moment of Inertia for composite shapes

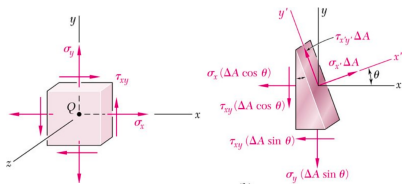
	$\bar{I}_x$	$A d^2$	$\bar{I}_x + A d^2$
A_1	$b_1 h_1^3 / 12$	$b_1 h_1 d_1^2$	$b_1 h_1 (d_1^2 + h_1^2 / 12)$
A_2	$b_2 h_2^3 / 12$	$b_2 h_2 d_2^2$	$b_2 h_2 (d_2^2 + h_2^2 / 12)$
A_3	$b_3 h_3^3 / 12$	$b_3 h_3 d_3^2$	$b_3 h_3 (d_3^2 + h_3^2 / 12)$

$$(\bar{I}_x)_i = (\bar{I}_x)_i + A_i d_i^2 \quad I_x = \sum (\bar{I}_x)_i$$

Symbols

- P axial force (N)
- T torque (N · m)
- L original length
- A cross-sectional area
- J polar MOI
- ρ distance from center
- σ normal stress
- τ shear stress
- ϵ axial strain
- δ elongation
- ϕ angle of twist (rads)
- E Young's modulus
- G shear modulus

Principle stresses and max shearing stresses



First Find $\sigma_x, \sigma_y, \& \tau_{xy}$

Second, Use angle to Transform to find $\sigma_{x'}, \sigma_{y'}, \tau_{x'y'}$

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) + \tau_{xy} \sin(2\theta)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) - \tau_{xy} \sin(2\theta)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin(2\theta) + \tau_{xy} \cos(2\theta)$$

Principal stresses (σ maximum and σ minimum)

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}, \text{ two angles separated by } 90^\circ$$

$$\sigma_{max,min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

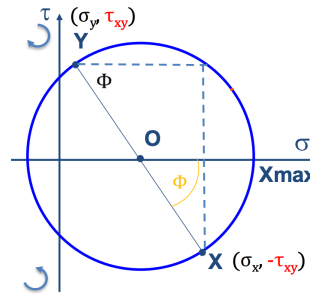
Max in-plane shearing stress

$$\tan(2\theta_s) = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}}, \text{ two angles separated by } 90^\circ$$

$$\tau_{max,min} = R = \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\text{max normal stress with inplane } \sigma' = \sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}$$

Moore Circle



Constructing 2D Mohr Circle:

Given: $\sigma_x, \sigma_y, \tau_{xy}$ (all > 0) at a particular orientation

1. Draw axes
2. Plot Point O, center, at $(\sigma_{avg}, 0)$
3. Plot Point X $(\sigma_x, -\tau_{xy})$, this corresponds to $\theta = 0^\circ$
4. Plot Point Y (σ_y, τ_{xy}) , this corresponds to $\theta = 90^\circ$
5. Draw a line through XY (through O): This is the diameter
6. Draw circle

$$\Phi = \tan^{-1} \frac{2\tau_{xy}}{\sigma_x - \sigma_y} (\Phi = 2\theta_p)$$

Constructing 3D Mohr Circle:

Given $\sigma_x, \sigma_y, \tau_{xy}$ and $\sigma_z = 0$ (unless given)

1. Find Average stress (σ_{avg}) and In-Plane Radius (R)
2. Find Principal Stresses 1 and 2 ($\sigma_{p1\&p2} = \sigma_{avg} \pm R$)
3. Determine Max & Min (σ_{max} & σ_{min}) from $\{\sigma_{p1}, \sigma_{p2}, \sigma_z\}$
4. Plot all the $\{\sigma_{p1}, \sigma_{p2}, \sigma_z\}$ values
5. Draw circles

$$\tau_{max} = \frac{1}{2} |\sigma_{max} - \sigma_{min}|$$

Parallel Axis Theorem

$$I = I_c + Ad^2$$

I : MOI for new axis

I_c : MOI for centroid

A : Cross section A

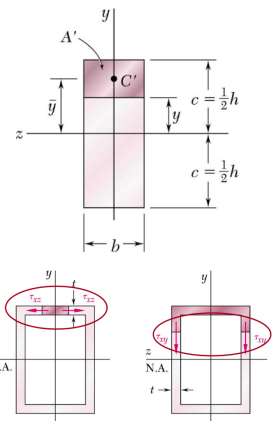
d : distance between parallel axis

Moment of Inertia

$$I_{\blacksquare} = \frac{1}{12} bh^3$$

$$I_o = \frac{\pi r^4}{4}$$

Shearing stress in beams



For rectangular

$$\tau_{xy} = \frac{3V}{2A} \left(1 - \frac{y^2}{c^2}\right)$$

$$\tau_{max} = \frac{3V}{2A}$$

Thin wall Shear

$$\tau = \frac{VQ}{It}$$

$$\text{Shear flow: } q = \frac{VQ}{I}$$

$$\text{Shearing Force: } F = q\ell,$$

Where ℓ : spacing between nails

V : internal shear force at section

Q : MOI above (or below) the point of interest, about the neutral axis

I : MOI of the entire cross-section about the neutral axis

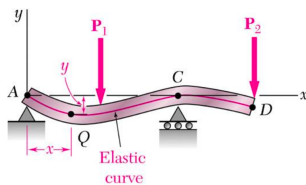
t : width of the cross-section at the point where τ is being calculated

$$1^{\text{st}} \text{ MOI: } Q = \sum A\bar{y}$$

$\bar{y}$: distance from shaded centroid to Neutral axis

A : Area of shaded region

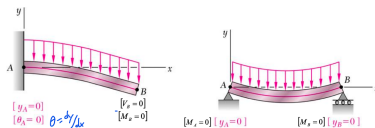
Beam Deflection



differential equation for the deflection

$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI} \quad \frac{d^3y}{dx^3} = \frac{V(x)}{EI} \quad \frac{d^4y}{dx^4} = \frac{-w(x)}{EI}$$

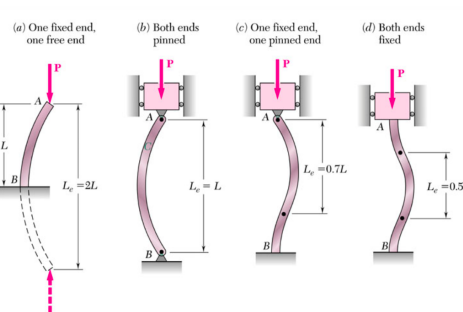
Boundary Conditions



Buckling of Slender Columns

$$P_{CR} = \frac{\pi^2 EI}{L_e^2}$$

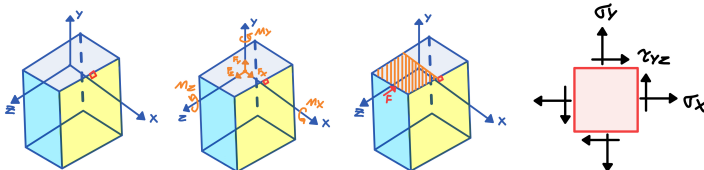
$$P_{allowable} = \frac{P_{CR}}{FS}$$



Lever Beams- Multi part

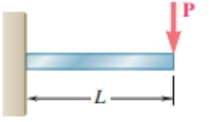
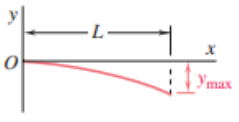
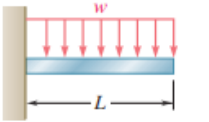
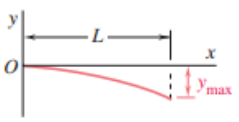
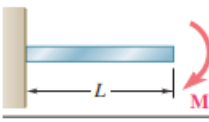
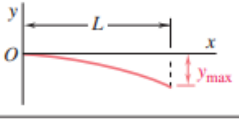
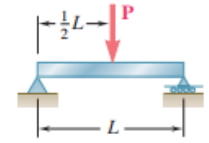
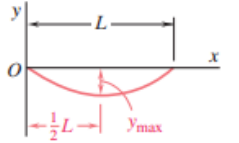
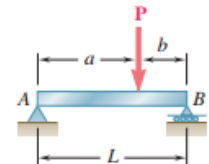
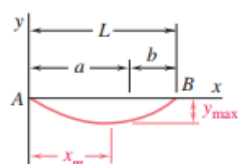
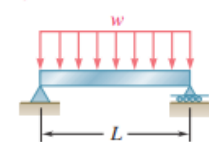
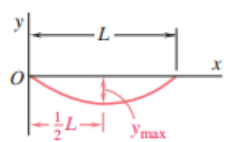
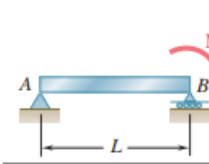
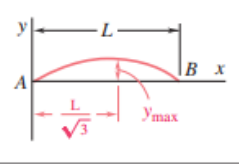
Step 1: Cut at the point we want to find shear around

Step 2: Find $F_x, F_y, F_z, M_x, M_y, M_z$, around the new coordinate frame



Step 3: $\sigma_x, \sigma_y, \tau_{xy}$: can be found many ways, use these 4 equations to find shear and normal in the new directions. Moments and forces anywhere can give shear and normal internal forces, add up these equations to find the shea and normal

$$\sigma_F = \frac{F}{A} \quad \sigma_M = \frac{My}{I} \quad \tau_F = \frac{FQ}{It} \quad \tau_M = \frac{Tc}{J}$$

Beam and Loading	Elastic Curve	Maximum Deflection	Slope at End	Equation of Elastic Curve
1 		$-\frac{PL^3}{3EI}$	$-\frac{PL^2}{2EI}$	$y = \frac{P}{6EI} (x^3 - 3Lx^2)$
2 		$-\frac{wL^4}{8EI}$	$-\frac{wL^3}{6EI}$	$y = -\frac{w}{24EI} (x^4 - 4Lx^3 + 6L^2x^2)$
3 		$-\frac{ML^2}{2EI}$	$-\frac{ML}{EI}$	$y = -\frac{M}{2EI} x^2$
4 		$-\frac{PL^3}{48EI}$	$\pm \frac{PL^2}{16EI}$	For $x \leq \frac{1}{2}L$: $y = \frac{P}{48EI} (4x^3 - 3L^2x)$
5 		For $a > b$: $-\frac{Pb(L^2 - b^2)^{3/2}}{9\sqrt{3}EIL}$ at $x_m = \sqrt{\frac{L^2 - b^2}{3}}$	$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL}$ $\theta_B = +\frac{Pa(L^2 - a^2)}{6EIL}$	For $x < a$: $y = \frac{Pb}{6EIL} [x^3 - (L^2 - b^2)x]$ For $x = a$: $y = -\frac{Pa^2b^2}{3EIL}$
6 		$-\frac{5wL^4}{384EI}$	$\pm \frac{wL^3}{24EI}$	$y = -\frac{w}{24EI} (x^4 - 2Lx^3 + L^3x)$
7 		$\frac{ML^2}{9\sqrt{3}EI}$	$\theta_A = +\frac{ML}{6EI}$ $\theta_B = -\frac{ML}{3EI}$	$y = -\frac{M}{6EIL} (x^3 - L^2x)$